To: The Budget Planning Committee

RE: Most Cost-Efficient Solution for Upgrading the Fleet

In light of ongoing public budget tightening, efficient resource allocation has become a critical priority for municipal decision-makers, and we propose a cost-minimization optimization model that balances financial constraints with service requirements. Detailed decision viables, objective functions and constraints of our approached have been identified as the following.

Decision Variablies

X = number of New Type A buses to purchase

Y = number of New Type B buses to purchase

Z = number of New Type C buses to retain

Objective function

Minimize Total Cost = 50,000X + 70,000Y

Constraints

1. Budget: 50,000X + 70,000Y <= 10,000,000

2. Seating Capacity: 25X + 50Y + 50Z >= 15,000

3. Drivers: X + Y + Z <= 450

4. Fuel efficiency: 10X + 8Y + 5Z / (X + Y + Z) >= 6

   Equivalent: 4X + 2Y – Z >=0

5. Non-negative: X,Y,Z >=0

Our Recommendation

We recommend to purchase (i) 67 Type A buses, (ii) 0 Type B buses and to retain (iii) 266 Type C buses, which achieves the lowest total cost of $ 3,350,000 after rounding to integers, according to the Excel Solver solution.

PLAN A: Cost-Efficency				
Bus type	**Seating capacity**	**Fuel efficiency [MPG]**	**Purchase cost [$]**	
New Type A bus	25.00	10.00	50000.00	
New Type B bus	50.00	8.00	70000.00	
Type C bus	50.00	5.00		
		New Type A buses	**New Type B buses**	**Type C buses**
DECISIONS VARIABLES	number of buses	66.67	0.00	266.67
OBJECTIVE FUNCTION	**budget**	3333333.333		
	Item	**Value**	**UPPER**	**LOWER**
CONSTRAINTS	budget	3333333.333	10,000,000	
	Fuel Efficiency	0		0
	Seating capacity	15000		15000
	drivers	333.3333333	450	